Fish Farming and Hatchery: Investment Appraisal Model

A fully linked Excel financial model that appraises a combined **fish farming and hatchery project** in Bangladesh. It answers one question a lender or investment committee would ask: *should this project be accepted?*

> **Illustrative case.** The company ("Sample Aqua Ltd."), its figures and all assumptions are a teaching and portfolio example. Nothing here is a real company's data, an offer, or investment advice.

Headline result (Base case)

Metric	Result
Project cost (fixed assets)	BDT 45.8 million
Bank term loan / owner equity	22.9 million / 25.3 million
Discount rate (WACC)	14.63%
NPV including recovery value	BDT 30.7 million
IRR including recovery value	29.3%
NPV, 5 years only (no recovery value)	BDT 3.1 million
IRR, 5 years only	16.6%
Payback / discounted payback	3.66 / 4.78 years
Profitability index	1.07
Minimum DSCR	1.61
Lending benchmarks met	8 of 8

Recommendation in the model: ACCEPT. NPV is positive and IRR is above the required return, and the project meets all eight lending benchmarks in the Base case.

What the headline depends on

- The 5-year NPV is positive but **thin (BDT 3.1 million)**. Most of the BDT 30.7 million headline comes from the **recovery value at the end of Year 5 (BDT 54.6 million)**: land, ponds and buildings, equipment, and the release of working capital.
- The recommendation therefore assumes the business continues or the assets keep their value after Year 5. Fish is a staple food, so a longer operating life is a reasonable view, but the model does not forecast beyond Year 5.
- Working-capital days (30 days receivables, 90 days inventory, 30 days payables) are **analyst assumptions** based on credit practice in the sector, not observed data.
- Under the **Downside** scenario the project is much weaker (see below). The model shows this rather than hiding it.

Scenarios

Scenario	Changes	NPV incl. recovery	Min DSCR	Benchmarks met
Base	None	30.7M	1.61	8 of 8
Upside	Price +5%, feed cost -5%	38.8M	1.84	8 of 8
Downside	Price -10%, yield -15%, feed +10%, capex +10%, interest +2 pp	6.1M	0.92	6 of 8

Switch scenarios with one selector on the Assumptions sheet. The Scenarios sheet shows all five years of revenue, EBITDA and net profit for each case. The Sensitivity sheet adds one-way sensitivities, a two-way NPV grid and break-even points.

What is in the workbook (19 sheets)

Group	Sheets	Purpose
Summary	Dashboard	One-page view: profitability, cash flow, working capital, investment metrics, owner and lender view
Inputs	Assumptions	All drivers in one place (blue cells): growth rates, tax, loan terms, working-capital days, scenario selector
Operating build-up	Farming, Hatchery, Operations, FA Schedule	Production, revenue, direct costs, overheads, fixed assets and depreciation
Working capital and funding	Working Capital, Funding	Receivables, inventory and payables; loan and equity split
Three statements	Income Statement, Cash Flow, Balance Sheet	Fully linked; Cash Flow includes a "Cash Flow Generated" section and a direct-method memo
Analysis	Segment Analysis, Investment Analysis, Owner Returns, Loan Analysis, Ratios	NPV, IRR, payback, DSCR, interest cover, returns to the owner, lending benchmarks
Risk	Scenarios and Risk, Sensitivity and Break-even	Scenario comparison, sensitivities, break-even
Control	Checks	15 integrity checks; the model reads "ALL OK" in the Base case

Key modelling logic

- **Revenue** = quantity x rate. Rates rise 5% a year; Year 1 rates are held in the Farming and Hatchery sheets (for example tilapia Tk 110 and carp Tk 180 per kg).
- **Costs:** feed +5%, salaries +5%, office and maintenance +10%, fuel +10%, other variable costs +5% a year. Festival bonus is two months of salary spread over twelve.
- **Year 2 expansion:** construction (BDT 2.0 million) and additional land (143 decimals at BDT 21,000, about BDT 5.0 million) are included in Year 2 capital expenditure.
- **Broodfish** are sold once, in Year 4: 60,000 tilapia at 0.5 kg and 5,000 carp at 5 kg, less 20% mortality (24,000 kg and 20,000 kg). The calculation is on the Assumptions sheet. Year 5 revenue is therefore lower than Year 4 (BDT 75.5 million against 83.1 million).
- **Working capital:** receivables = revenue x DSO / 365, inventory from cost of sales x DIO / 365, payables from purchases x DPO / 365.
- **Funding:** loan = 50% of fixed assets (BDT 22.9 million), 5-year tenor, 1-year grace, repaid in Years 2 to 5, interest 12% on the opening balance.
- **Tax:** 27.5% with loss carry-forward. Minimum cash BDT 0.5 million; dividend payout 60% of profit.
- **WACC** = 47.5% x 8.7% (after-tax cost of debt) + 52.5% x 20% (cost of equity) = 14.63%.
- **Colour code:** blue = input, black = formula, green = link to another sheet.
- **Units:** statements in BDT '000; Farming, Hatchery, Fixed Assets and Operations in full Taka; Dashboard in BDT million.

How to use

1. Open the workbook in Excel (or LibreOffice) and enable calculation.
2. Start at **Dashboard**, then read **Assumptions**.
3. Change only **blue input cells**. Everything else recalculates.
4. Pick a scenario on the Assumptions sheet and watch Dashboard, Scenarios and Risk, and Checks.
5. Confirm the **Checks** sheet shows "ALL OK" after any change.

Verification

- 15 in-workbook integrity checks (balance sheet balances, cash ties to the cash flow, month-wise revenue equals annual revenue, direct and indirect methods agree, and more).
- An independent Python re-computation reproduced revenue, EBITDA, free cash flow, WACC, NPV and IRR.
- The workbook was rebuilt and recalculated end to end, so stored values match the formulas.

Limitations

- Five explicit forecast years; value after Year 5 is a recovery-value assumption, not a forecast.
- Production is not capacity-constrained by formula. Stocking per decimal of pond is an input, so raising it without adding pond area overstates revenue.
- Price, yield, mortality and disease risk are tested only through the scenario and sensitivity sheets.
- Credit terms, tax and interest rates reflect the assumptions stated on the Assumptions sheet and should be updated before real use.

Skills demonstrated

Financial modelling, three-statement integration, working-capital and cash-flow analysis, capital budgeting (NPV, IRR, payback, PI), WACC, debt-service and lending-benchmark analysis, scenario and sensitivity analysis, model auditing and independent verification.

Files

- `Sample_Aqua_Fish_Project_Model_Farming_Hatchery.xlsx`: the model
- `README.md` / `README.pdf`: this document

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